

COURSE 3 - DATA ENGINEERING 2026**C3-104AR - EnergyOps Gate Review**

Protected. Open only after the matching Gate attempt is saved.

Artifact	C3-104AR
Status	2026 - LEARNER EDITION
Phase	C3-09 Production Capstone & Engineering Transition 15 lessons 36-48 focused hours
Prerequisite	C3-08 Gate PASS; all earlier C3 evidence retained.
Assessment boundary	C3-104 is the phase Gate only. C3-113 remains the later course-wide Final Production Gate.
Brand	Charcoal #1F2937 Teal #14B8A6 Soft Blue #3B82F6 AMOLED #000

House teaching rule: requirement -> mental picture -> small proof -> guided integrated build -> expected result -> why it worked -> break/fix -> changed independent case -> validation -> production reasoning -> evidence. Integration is not mastery until the learner can explain and recover the system independently.

Competency diagnosis

Evaluate A-J separately. A failed Gate enters targeted remediation. A fresh full B or C is opened only by explicit assignment; targeted repair never automatically exposes the next full retest.

Strong evidence

- Requirements justify the architecture and unnecessary complexity is rejected.
- Correction/new records converge under replay; stable key/version rule is explicit.
- Serving result reconciles independently to accepted source/current state.
- Backfill/partial rerun has bounded blast radius.
- Spark/distributed result has parity against deterministic reference.
- Quality gate catches silent corruption even when row counts/totals alone could mislead.
- Incident proof includes containment, repair, reprocessing, validation and prevention.
- Security uses least privilege and sanitized evidence; live deny/allow is only claimed if executed.
- CI/CD cannot deploy through failed required controls and includes post-deploy data validation.
- README/runbook/evidence support an independent architecture defense.